

CHAPTER 13

MORPHOLOGICAL IMAGE PROCESSING

WHAT WILL WE LEARN?

- What is mathematical morphology and how is it used in image processing?
- What are the main morphological operations and what is the effect of applying them to binary and grayscale images?
- What is a structuring element (SE) and how does it impact the result of a morphological operation?
- What are some of the most useful morphological image processing algorithms?

What is mathematical morphology?

- Provide tools for representing, describing, and analyzing **shapes** in images
- Extract geometrical and topological information through transformations using *structuring element*

FUNDAMENTAL CONCEPTS AND OPERATIONS

- Set theory and set operations:

- Union ($\cup$)
- Intersection ($\cap$)
- Complement
- Difference of two sets A and B
- Translation of set A by point w is denoted by A_w and defined as

$$A_w = \{c \mid c = a + w, \text{ for } a \in A\}$$

- The reflection of set A relative to the origin of a coordinate system is defined as

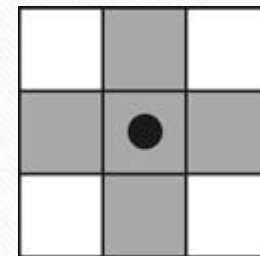
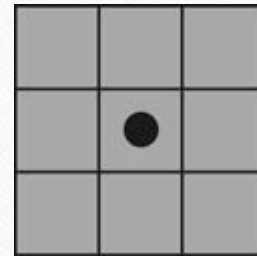
$$\hat{A} = \{z \mid z = -a, \text{ for } a \in A\}$$

- Matlab expression

- OR ($|$)
- AND ($\&$)
- NOT ($\sim$)
- ($A \& \sim B$)

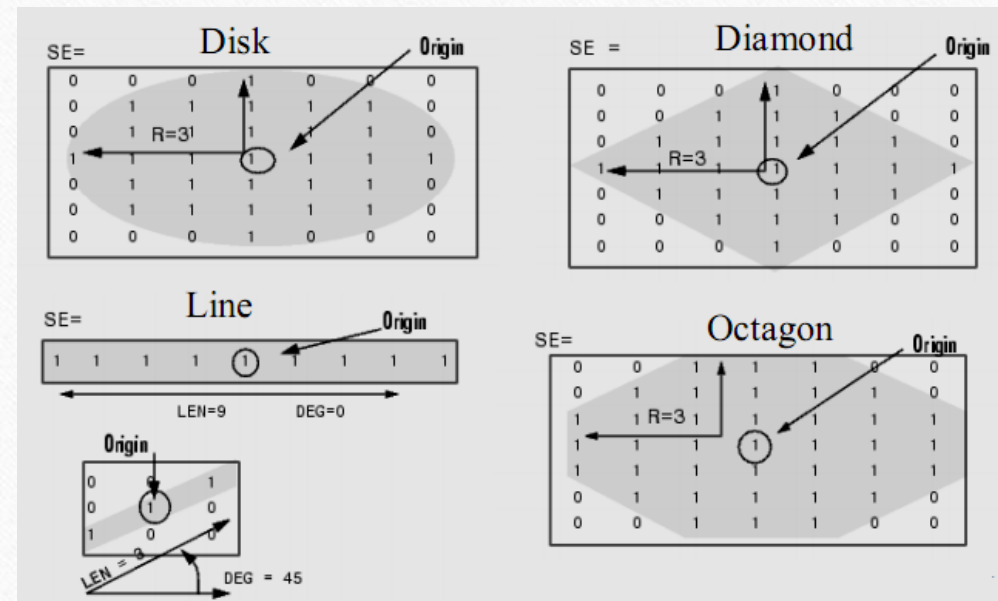
The Structuring Element (SE)

- The structuring element is the basic neighborhood structure associated with morphological image operations
- The black dot corresponds to their origin (reference point), the gray squares represent 1 (true), and the white squares represent 0 (false).



In MATLAB

- **strel**: for creating structuring elements which supports arbitrary shapes, as well as commonly used ones, such as square, diamond, line, and disk.

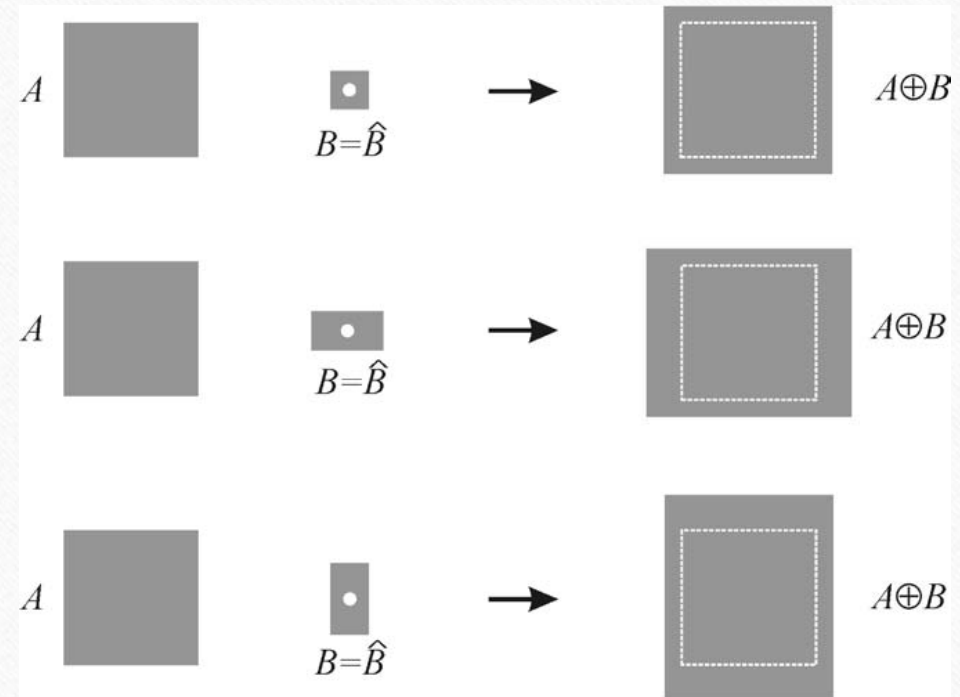


DILATION AND EROSION

- Dilation is a morphological operation whose effect is to “grow” or “thicken” objects in a binary image

$$A \oplus B = \{z | (\hat{B})_z \cap A \neq \emptyset\}$$

- In Matlab: **imdilate**

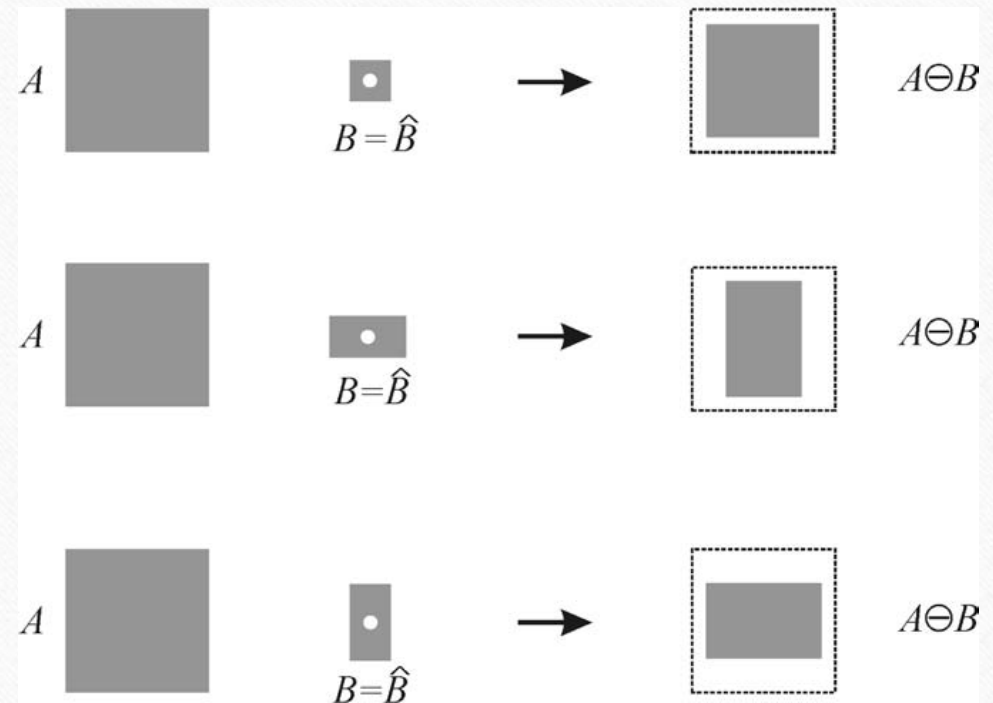


DILATION AND EROSION

- Erosion is a morphological operation whose effect is to “shrink” or “thin” objects in a binary image

$$A \ominus B = \{z | (\hat{B})_z \cup A^c \neq \emptyset\}$$

- In Matlab: **imerode**

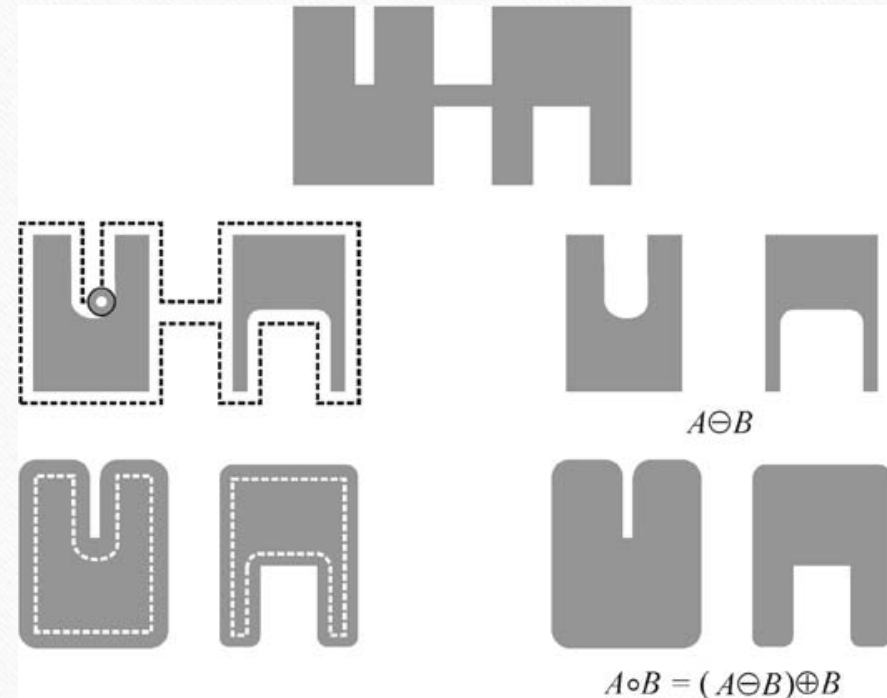


COMPOUND OPERATIONS

- **Opening:** $A \circ B$, is the erosion of A by B followed by the dilation of the result by B .

$$A \circ B = (A \ominus B) \oplus B$$

- Opening is typically used to remove thin protrusions from objects and to open up a gap between objects connected by a thin bridge without shrinking the objects

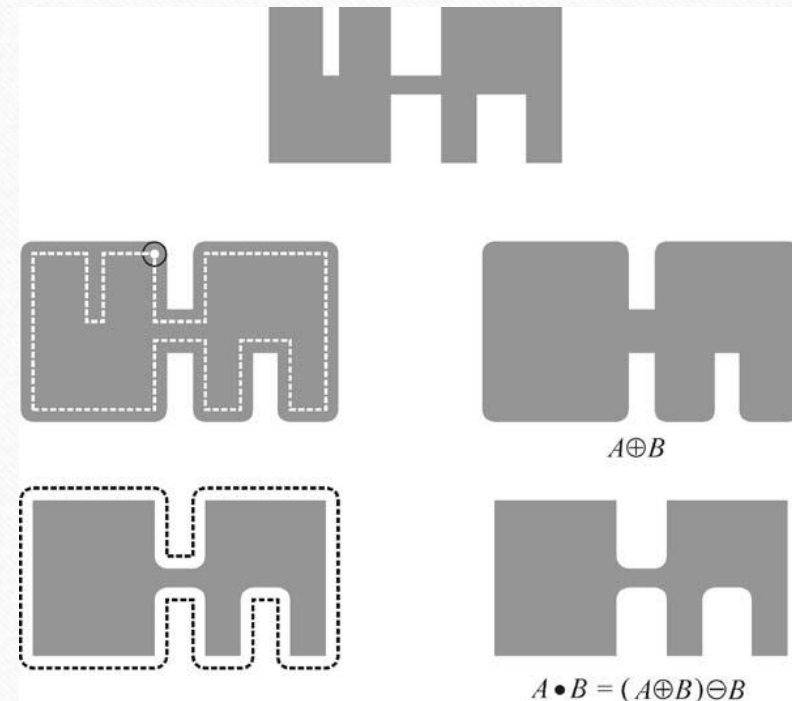


COMPOUND OPERATIONS

- **Closing:** is the dilation of A by B followed by the erosion of the result by B

$$A \bullet B = (A \oplus B) \ominus B$$

- Closing is used to fill small holes, fuse narrow breaks, and close thin gaps in the objects within an image

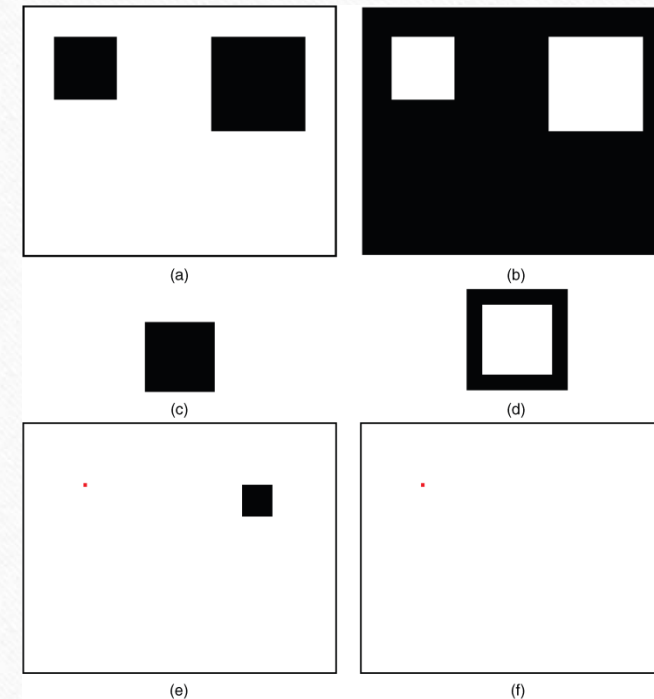


COMPOUND OPERATIONS

- **Hit-or-Miss Transform** : uses two structuring elements (B_1 and B_2) designed in such a way that the output image will consist of all locations that match the pixels in B_1 (a hit) and that have none of the pixels in B_2 (a miss)

$$A \otimes B = (A \ominus B_1) \cap (A^c \ominus B_2)$$

- HoM transform can be used to locate squares of a certain size in a binary image



MORPHOLOGICAL FILTERING

- Morphological filters are Boolean filters that apply a many-to-one binary (or Boolean) function h within a window W in the binary input image $f(x, y)$, producing at the output an image $g(x, y)$ given by

$$g(x, y) = h [Wf(x, y)]$$

- OR: is equivalent to a morphological dilation with a square SE of the same size as W .
- AND: is equivalent to a morphological erosion with a square SE of the same size as W .
- MAJ (Majority): is equivalent to a median filter applicable to binary images.

Boundary Extraction

- Dilation, erosion, and set difference operations can be combined to perform boundary extraction of a set A , denoted by $BE(A)$

- Internal Boundary: Consists of the pixels in A that sit at the edge of A

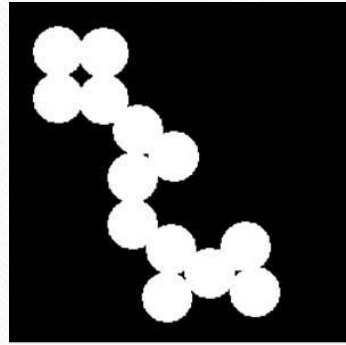
$$BE(A) = A - (A \ominus B)$$

- External Boundary: Consists of the pixels outside A that sit immediately next to A

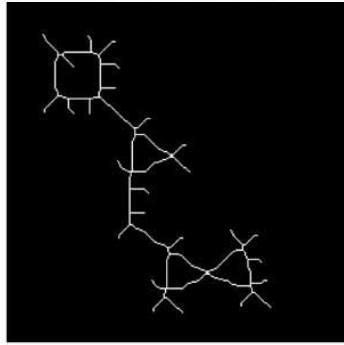
$$BE(A) = (A \oplus B) - A$$

- Morphological Gradient: Consists of the combination of internal and external boundaries.

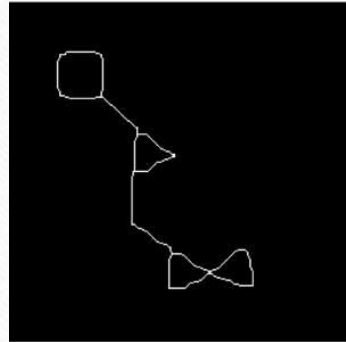
$$BE(A) = (A \oplus B) - (A \ominus B)$$



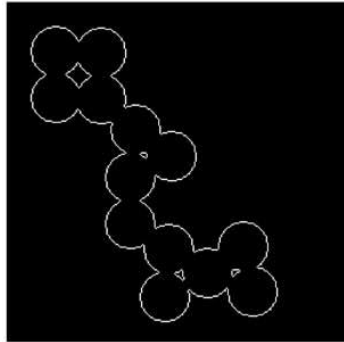
(a)



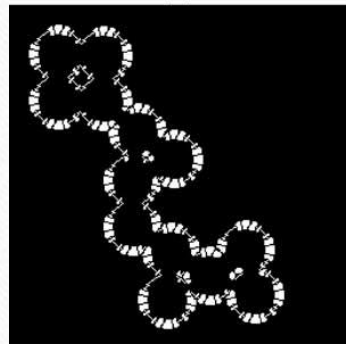
(b)



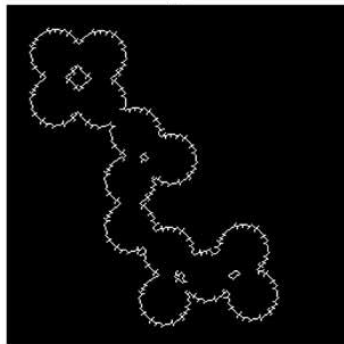
(c)



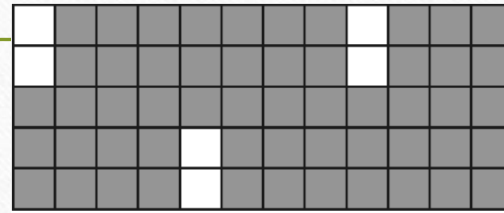
(d)



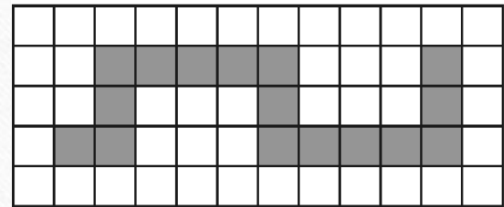
(e)



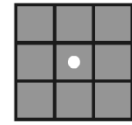
(f)



A
(a)

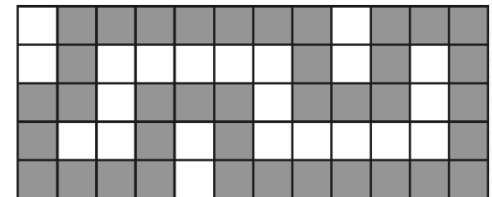


$A \oplus B$
(c)



B

(b)



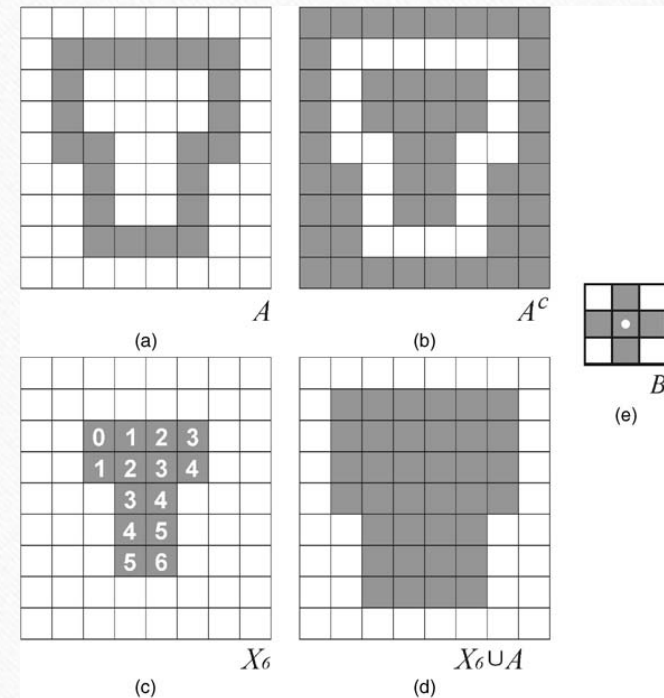
$A - (A \oplus B)$
(d)

Region Filling

- fill up the entire region with 1s using p as a starting point (i.e., setting it as 1).

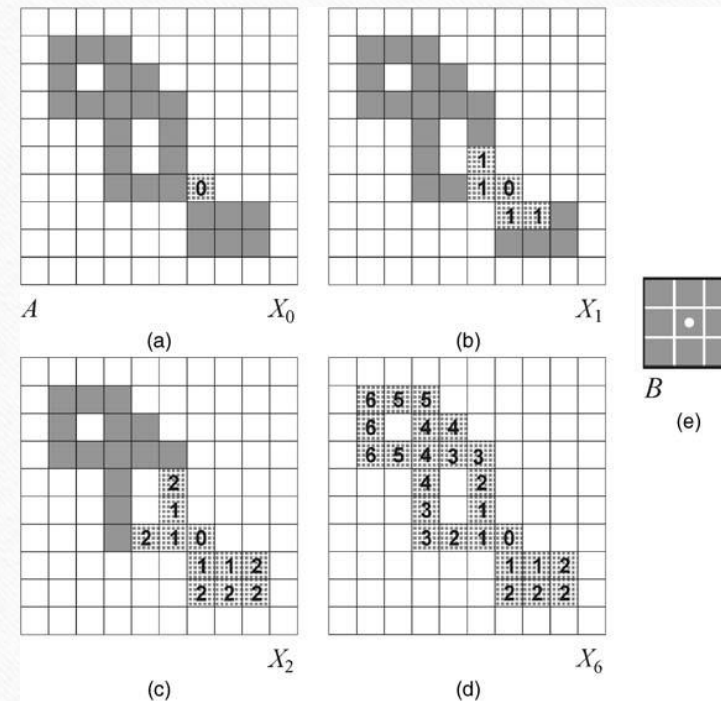
- Using an iterative procedure:

$$X_k = (X_{k-1} \oplus B) \cap A^c, k = 1, 2, 3, \dots$$



Extraction and Labeling of Connected Components

- In Matlab: `bwlabel`
- See Tutorial 13.2



GRAYSCALE MORPHOLOGY

- Morphological operations can be extended to grayscale images.
- The mathematical formulation of grayscale morphology uses an input image $f(x,y)$ and a structuring element (SE) $b(x,y)$.
 - Structuring elements in grayscale morphology: *nonflat* and *flat*.
- In MATLAB: **strel**
 - Nonflat SEs can be created with the same function used to create flat SEs, but we must also pass a second matrix (containing the height values) as a parameter.

GRAYSCALE MORPHOLOGY

- **Dilation:**
- The dilation of an image $f(x,y)$ by a flat SE $b(x,y)$ is defined as:

$$[A \oplus B](x, y) = \max \{f(x + s, y + t) | (s, t) \in D_b\}$$

where D_b is called the domain of b .

- For a nonflat SE, $b_N(x,y)$:

$$[A \oplus B](x, y) = \max \{f(x + s, y + t) + b_N(s, t) | (s, t) \in D_b\}$$

GRAYSCALE MORPHOLOGY

- Erosion

- The erosion of an image $f(x,y)$ by a flat SE $b(x,y)$ is defined as:

$$[A \ominus B](x, y) = \min \{ f(x + s, y + t) | (s, t) \in D_b \}$$

where D_b is called the domain of b .

- For a nonflat SE, $b_N(x,y)$:

$$[A \ominus B](x, y) = \min \{ f(x + s, y + t) - b_N(s, t) | (s, t) \in D_b \}$$

GRAYSCALE MORPHOLOGY

- Example 13.8



(a)



(b)



(c)

GRAYSCALE MORPHOLOGY

- Opening :

$$f \circ b = (f \ominus b) \oplus b$$

- Closing

$$f \bullet b = (f \oplus b) \ominus b$$

GRAYSCALE MORPHOLOGY

- Example 13.9



(a)



(b)



(c)



(d)



(e)



(f)

GRAYSCALE MORPHOLOGY

- Top-hat transformation

$$\text{Top-hat}(f) = f - (f \circ b)$$

- Bottom-hat transformation

$$\text{Bottom-hat}(f) = (f \bullet b) - f$$

- In MATLAB: **imtophat** and **imbothat**

Top-hat and bottom-hat

- Example 13.10



(a)



(b)

Hands-on

- Tutorial 13.1: Binary morphological image processing (page 325)
- Tutorial 13.2: Basic morphological algorithms (page 330)